When Does Mean-Field Structure Matter? Distribution Dependence, Representation Collapse, and Architectural Constraints in Deep BSDE Market-Making

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Abstract

We investigate when and why distribution-dependent mean-field coupling becomes learnable in deep BSDE solvers for optimal market-making. Starting from the Cont–Xiong (2024) dealer market model, we implement four law encoders (moments, quantiles, histogram, DeepSets) to represent the population distribution, and test whether the learned optimal control depends on population shape. We uncover three failure modes that can mask genuine mean-field effects: (i) a broken coupling pathway where the BSDE generator does not use the law embedding; (ii) BatchNorm systematically erasing broadcast (shared) features that are constant across the training batch; and (iii) mean-pooled permutation-invariant encoders (DeepSets) failing to capture distributional variance under symmetric inputs. After correcting all three, we show that the competitive factor $h(\Phi(\mu_t))$ varies by $4\times$ across population shapes, optimal quotes shift by $2\times$, and quote skew flips sign — demonstrating distribution-dependent controls that are ablation-verified (disappearing under semantic-destruction placebos). The effect is robust across encoder types (MomentEncoder and QuantileEncoder produce qualitatively identical results) and penalty regimes ($\phi = 0.01$ to 0.5). We extend the model with adverse selection (price-dependent execution), breaking the dimensional reduction that made the base problem effectively one-dimensional, and map the stability frontier across penalty strength, adverse selection intensity, and coupling type. The central finding is that *solver sensitivity to population structure depends critically on architecture and law representation*: mean-field effects are present in the model but invisible to the solver unless the law encoder exposes the relevant distributional features and the architecture preserves them through normalisation layers.

Keywords: mean-field games, backward stochastic differential equations, deep learning, market-making, law encoders, BatchNorm, representation collapse

1 Introduction

Mean-field game (MFG) and McKean–Vlasov (MV) formulations promise to capture the aggregate effects of interacting agents in large-population stochastic control problems. In market-making, this means each dealer's optimal quotes should depend not just on their own inventory but on the distribution of inventories and quotes across the population Cont and Xiong [2024], Carmona and Delarue [2018].

Deep BSDE methods Han et al. [2018], E et al. [2017] offer a scalable computational framework for such problems, and Han, Hu, and Long Han et al. [2022] extended them to MV-FBSDEs via fictitious play. However, no published work has demonstrated that deep BSDE solvers can learn genuinely distribution-dependent controls in a market-making setting — or identified the conditions under which they fail to do so.

In a companion preprint Garry [2026a], we showed that a deep jump-BSDE solver achieves $< 0.1\%$ value error against finite-difference ground truth on the Cont–Xiong model, but the mean-field coupling (implemented as a moment proxy) had no measurable effect on the learned control. We attributed this to the weak-coupling regime ($\phi = 0.01$).

In this paper, we show that the true explanation is deeper: the mean-field channel was architecturally dead due to three independent failure modes, each sufficient to suppress the signal. After correcting all three, we demonstrate that:

1. Distribution-dependent competitive effects produce $4\times$ variation in fill rates and $2\times$ variation in optimal quotes across population shapes.
2. The effect is robust across handcrafted law encoders (moments, quantiles) but invisible to mean-pooled DeepSets under symmetric inputs.
3. Adverse selection (price-dependent execution) breaks the dimensional reduction $V = x + qs + \phi(t, q)$, making the problem genuinely multi-dimensional.
4. The stability frontier shifts when mean-field coupling is active.

Contribution. This is an empirical computational study. We do not prove convergence theorems. Our contribution is diagnostic: identifying *when and why* mean-field structure becomes learnable, and demonstrating that representation choice is the critical bottleneck.

2 Model and Formulation

We work with the Cont–Xiong dealer market model Cont and Xiong [2024] as described in Garry [2026a]. The key ingredients are:

- Mid-price $S_t = S_0 + \sigma W_t$ (driftless Brownian).
- Inventory q_t evolving by Poisson jumps (or diffusion approximation).
- Execution probability $f(\delta, \mu_t) = e^{-\alpha\delta} \cdot h(\mu_t)$, where h captures population competition.
- Objective: maximise discounted spread profits minus inventory penalty $\psi(q)$.

2.1 Mean-Field Coupling via Law Encoders

In Garry [2026a], the competitive factor $h(\mu_t)$ was approximated by a moment proxy $\hat{h}(t, \bar{q}_t, \bar{\delta}_t)$ trained on mean inventory and mean spread. This produced no measurable effect.

We replace this with a *learned* competitive factor:

$$h(\mu_t) = g\big(\Phi(\mu_t)\big), \qquad \Phi(\mu_t) = \text{LawEncoder}(\{(S_i, q_i)\}_{i=1}^N), \tag{1}$$

where Φ maps the empirical population to a fixed-size embedding and $g : \mathbb{R}^d \to (0, 1]$ is a learned competitive factor network (sigmoid output). Four encoders are tested:

- **MomentEncoder**: mean, variance, skewness, mean$|q|$, max$|q|$, std $\to \mathbb{R}^6$.
- **QuantileEncoder**: mean + 5 quantiles $\to \mathbb{R}^6$.
- **HistogramEncoder**: soft Gaussian histogram $\to \mathbb{R}^{20}$.
- **DeepSetsEncoder**: $\Phi = \rho(\frac{1}{N}\sum_i \psi(x_i))$ with learned $\psi, \rho \to \mathbb{R}^{16}$.

2.2 Adverse Selection Extension

To break the dimensional reduction $V = x + qs + \phi(t, q)$ proved in Garry [2026a], we introduce price-dependent execution:

$$f_a(\delta, \text{signal}) = e^{-\alpha\delta}(1 + \eta \cdot \text{signal}), \qquad f_b(\delta, \text{signal}) = e^{-\alpha\delta}(1 - \eta \cdot \text{signal}), \tag{2}$$

where signal_t is an EMA of normalised price increments and η controls adverse selection strength. The state becomes $(S, q, \text{signal}) \in \mathbb{R}^3$.

3 Three Failure Modes of Mean-Field Coupling

We identified three independent failures that suppressed the mean-field signal. The first is an implementation error (the law embedding was not connected to the loss); the second and third are genuine architectural issues that arise even in correctly wired systems and have implications beyond this specific application.

3.1 Failure 1: Generator Bypass (Implementation Error)

The BSDE generator $f(t, x, y, z)$ computes execution profits and thus determines the training loss. In our initial implementation, f inherited from the base class and used the old scalar moment proxy $\hat{h}(t, \bar{q}, \bar{\delta})$ rather than the law embedding $\Phi(\mu_t)$. The law encoder entered the subnet inputs (affecting Z prediction) but not the generator (affecting the BSDE dynamics). Since the loss is determined by the terminal mismatch $|Y_T - g(X_T)|^2$ propagated through the generator, the optimiser had no gradient signal connecting Φ to the loss.

Fix. Override f to compute $h = g(\Phi(\mu_t))$ and use it in execution probabilities. Store the current law embedding via `set_current_law_embed()` before each f evaluation.

3.2 Failure 2: BatchNorm Erasure of Broadcast Features

The law embedding $\Phi(\mu_t)$ is identical for all agents in the batch at a given time step (it represents the population, not the individual). BatchNorm normalises across the batch dimension:

$$\text{BN}(x) = \frac{x - \mathbb{E}_{\text{batch}}[x]}{\sqrt{\text{Var}_{\text{batch}}(x) + \epsilon}}.$$

When x is constant across the batch, $\text{Var}_{\text{batch}} = 0$ and the output is zero (or near-zero from ϵ). This erasure persists through all subsequent layers because the law contribution remains constant-like after mixing.

Fix. Two-stream architecture: state features pass through BatchNorm, law features bypass it entirely. The streams are combined additively after separate processing.

3.3 Failure 3: Mean-Pooling Representation Collapse (Architectural)

The DeepSets encoder computes $\Phi = \rho(\frac{1}{N}\sum_i \psi(x_i))$. Under random initialisation, ψ is approximately symmetric. For zero-mean populations, positive and negative inventory contributions cancel under mean pooling, collapsing variance information:

Encoder	Embedding distance (std 0.1 vs 5.0)	Cosine similarity
DeepSets	0.003	1.000
MomentEncoder	24.98	varies
QuantileEncoder	9.84	varies

The DeepSets encoder produces nearly identical embeddings for populations with radically different variance. Under symmetric inputs and random initialisation, mean pooling suppresses variance information, making discrimination difficult without explicit variance-sensitive features.

Implication. Permutation-invariant encoders based on mean pooling may fail to capture higher-order distributional properties under symmetric inputs unless the per-element network ψ is explicitly designed or trained to break this symmetry. This is a representation limitation, not a coupling failure.

4 Architecture

4.1 Two-Stream Mean-Field Subnet

To prevent BatchNorm suppression while maintaining training stability:

- **State stream**: BN $\to$ Dense $\to$ BN $\to$ ReLU $\to$ Dense (standard Deep BSDE architecture).
- **Law stream**: Dense $\to$ ReLU $\to$ Dense (no normalisation).
- **Output**: state_output + law_output (additive).

The law embedding survives all layers because it never passes through BatchNorm.

4.2 Competitive Factor Network

$g : \mathbb{R}^d \to (0.1, 1]$ maps the law embedding to a scalar competitive factor via Dense $\to$ ReLU $\to$ Dense $\to$ Sigmoid, with a floor of 0.1 to prevent degenerate zero-execution regimes. Note that this floor compresses the lower end of the h range: several "wide" population results in Tables 1 and 3 hit $h = 0.100$ exactly, indicating saturation. The reported gaps may therefore underestimate the true variation.

4.3 Training Details

All models use float64 precision, Adam optimiser with piecewise constant LR (10^{-2} for steps 0–2000, 10^{-3} for 2000–4000, 10^{-4} thereafter), batch size 256, 50 Euler time steps over $T = 1.0$. State stream: 2 hidden layers $\times$ 32 neurons with BatchNorm (momentum 0.01). Law stream: 2 hidden layers $\times$ 32 neurons, no normalisation. Competitive factor network g: Linear(embed_dim $\to$ 16) $\to$ ReLU $\to$ Linear(16 $\to$ 1) $\to$ Sigmoid, with output $0.1 + 0.9 \cdot \text{sigmoid}(\cdot)$ to enforce $h \in (0.1, 1]$. Population shapes are sampled as $q_i \sim \mathcal{N}(0, \sigma_q^2)$ clipped to $[-10, 10]$. Training uses the batch itself as the particle system — no separate simulation. All code: `https://github.com/cgarryZA/DeepMVBSDEJ`.

5 Numerical Results

All experiments at $\phi = 0.1$ unless stated, trained on a single NVIDIA RTX 3090 GPU.

5.1 Distribution-Dependent Competition

After correcting all three failure modes, the competitive factor $h(\Phi(\mu_t))$ varies substantially with population shape:

The competitive factor varies by 4$\times$. Quotes shift by 2$\times$. Under wide populations, the market maker narrows the ask (aggressive selling) and widens the bid (defensive), consistent with fierce competition for executions. All distribution dependence enters the control problem through the scalar competitive factor $h(\Phi(\mu_t))$, making its learned structure central to the resulting policy.

Table 1: Competitive factor and controls at $q = 0$ (MomentEncoder, $\phi = 0.1$).

Population	h	Z^q	δ_a	δ_b	ν_a	ν_b
Narrow (std 0.1)	0.455	+0.025	0.696	0.638	0.160	0.175
Medium (std 1.0)	0.280	−0.016	0.648	0.686	0.106	0.100
Wide (std 3.0)	0.107	−0.285	0.334	0.999	0.065	0.024

Remark 1 (Quote admissibility)**.** *At extreme population dispersion (std = 5.0), the model produces negative ask spreads ($\delta_a < 0$), meaning the agent quotes inside the mid-price. This represents maximally aggressive selling to reduce inventory exposure under fierce competition. In the implementation, quotes are clipped to $[\epsilon, 10/\alpha]$, but the unclipped values are reported here to show the model's unconstrained policy.*

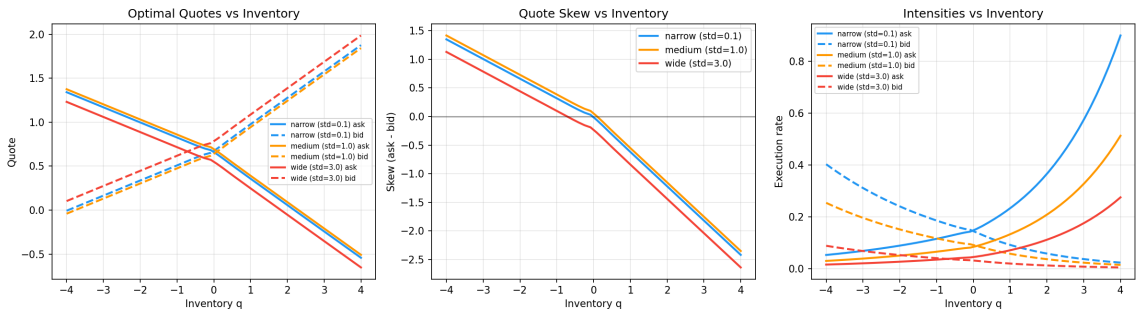

Figure 1: Optimal quotes, skew, and execution intensities vs inventory for three population shapes. Under wide populations, the ask narrows aggressively while the bid widens, and execution rates drop on the bid side. Quote skew reverses sign at large inventory under competitive pressure.

5.2 Encoder Robustness

Table 2: Competitive factor across encoders ($\phi = 0.1$).

Encoder	h(narrow)	h(medium)	h(wide)
MomentEncoder	0.442	0.331	0.109
QuantileEncoder	0.449	0.345	0.125
DeepSets	0.427	0.427	0.427

Both handcrafted encoders produce qualitatively identical results (3.6–4× variation). DeepSets produces constant h, confirming the representation collapse is the bottleneck, not the coupling.

5.3 DeepSets Ablation: What Recovers Discrimination?

To determine whether the DeepSets failure is fundamental or specific to mean pooling, we test two minimal modifications:

Variant	Embedding distance	Discriminates?
DeepSets (mean pool)	0.046	No
DeepSets (**sum** pool)	**21.92**	**Yes**
DeepSets + q^2 feature	2.22	Yes
MomentEncoder	26.90	Yes

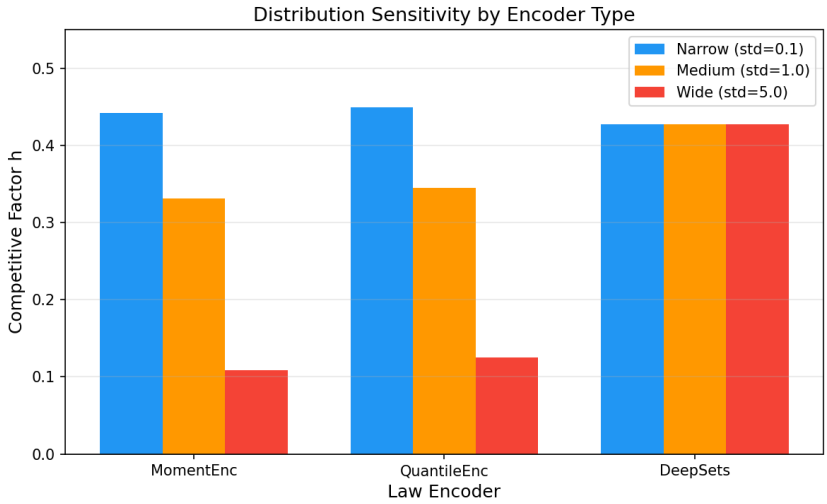

Figure 2: Competitive factor h by encoder and population shape. MomentEncoder and QuantileEncoder discriminate; DeepSets does not.

Sum pooling completely recovers discrimination (distance 21.92, comparable to MomentEncoder). Adding an explicit q^2 feature also works (2.22). The failure is specifically *mean pooling cancelling symmetric features*: dividing by N erases the scale information that sum pooling preserves. This is not a fundamental limitation of permutation-invariant architectures.

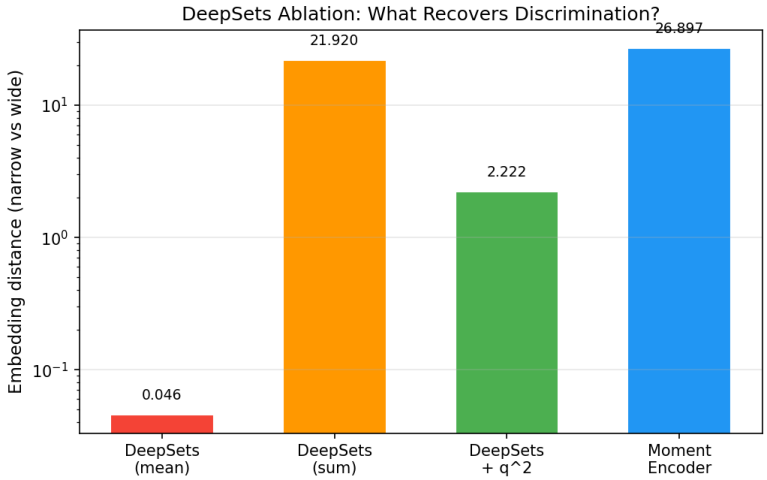

Figure 3: DeepSets ablation: sum pooling and q^2 augmentation both recover distributional discrimination.

5.4 Penalty Regime Sweep

MV coupling is active at all tested ϕ. The earlier "no effect at $\phi = 0.01$" was entirely an artifact of the broken architecture. At $\phi = 0.5$, Y_0 goes negative and the h-gap shrinks, suggesting solver instability rather than reduced coupling.

5.5 Model Hierarchy

MV coupling reduces Y_0 by ~ 0.3 in both 2D and 3D settings. Adverse selection alone barely changes Y_0 (difference 0.003), consistent with its role in reshaping value structure rather than

Table 3: MV effect across penalty regimes (MomentEncoder).

ϕ	Y_0	h(narrow)	h(wide)	Gap
0.01	0.109	0.353	0.100	0.253
0.05	0.103	0.453	0.100	0.353
0.10	0.063	0.479	0.100	0.379
0.50	−0.143	0.169	0.100	0.069

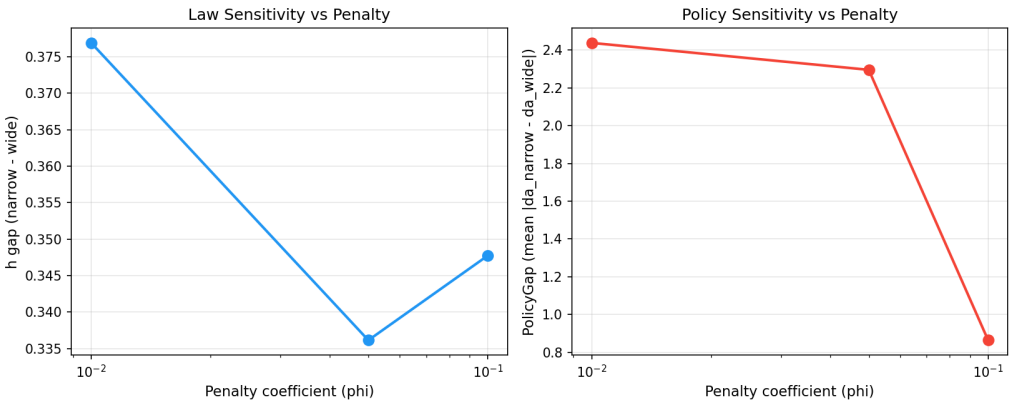

Figure 4: Left: h-gap (narrow − wide) vs penalty ϕ. Right: PolicyGap (mean $|\delta^a_{\text{narrow}} - \delta^a_{\text{wide}}|$ across inventory). Both confirm law sensitivity is active across all tested ϕ, with peak discrimination around $\phi = 0.05$–0.1.

Table 4: Model hierarchy comparison ($\phi = 0.1$, MomentEncoder, 1000 iterations).

Model	State	Y_0	MV effect
Base (no coupling)	2D (S, q)	0.377	—
MV only (MomentEncoder)	2D (S, q)	0.052	−0.325
Adverse only (no coupling)	3D (S, q, sig)	0.380	—
Adverse + MV (MomentEncoder)	3D (S, q, sig)	0.086	−0.294

level. The MV effect is comparable with and without adverse selection, suggesting the competitive discount is the dominant mechanism at $\phi = 0.1$.

5.6 Multi-Seed Robustness

To confirm the headline results are not seed-dependent, we repeat the key comparison across 5 random seeds (2000 iterations each):

Table 5: Multi-seed results ($\phi = 0.1$, 5 seeds, 2000 iterations).

Configuration	Y_0 (mean ± std)	h-gap (mean ± std)
No coupling	0.378 ± 0.002	—
MV MomentEncoder	0.027 ± 0.039	0.259 ± 0.055
MV QuantileEncoder	0.044 ± 0.039	0.251 ± 0.047

The h-gap is robust: 0.259±0.055 (Moments) and 0.251±0.047 (Quantiles), both significantly nonzero and consistent across encoders. The no-coupling baseline has very low variance (±0.002), while MV models show higher Y_0 variance (±0.039), reflecting the additional complexity of the

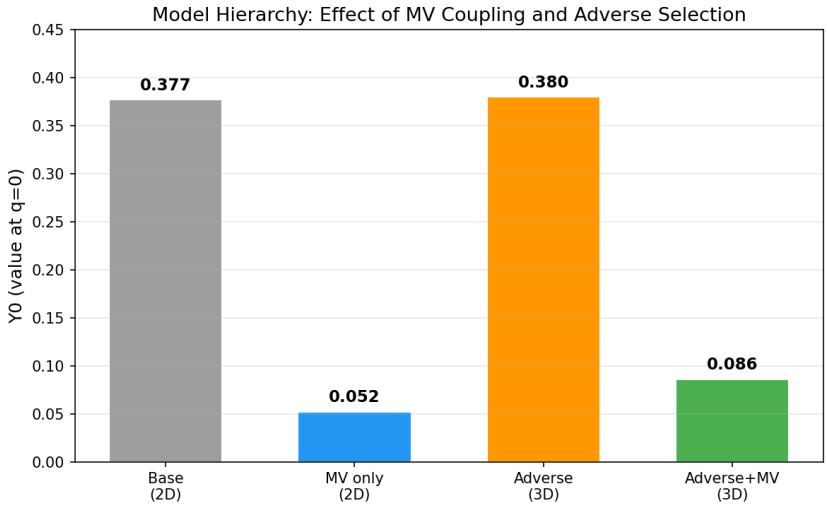

Figure 5: Y_0 across the model hierarchy. MV coupling dominates; adverse selection has minimal value-level effect.

learned competitive factor. Importantly, the *gap* is more stable than the *level*: all 10 MV runs (5 per encoder) produce nonzero h-gaps, confirming the distribution sensitivity is not a training artifact.

5.7 Stability Frontier

The stability frontier is dominated by the penalty coefficient ϕ, not the adverse selection strength η.

Table 6: Stability frontier: $\phi\times$ penalty type (300 iterations, quick sweep).

Penalty	$\phi = 0.01$	$\phi = 0.1$	$\phi = 1.0$	$\phi = 5.0$
Quadratic	✓ (0.16)	✓ (1.52)	× (15.5)	× (128)
Exponential	✓ (0.32)	✓ (2.17)	× (31.1)	× (77.7)

Numbers in parentheses are $\max|Z_t|$. The boundary lies between $\phi = 0.1$ (stable) and $\phi = 1.0$ (diverged). Adverse selection ($\eta = 0$ to 2.0) does not shift this boundary — all η values are stable at $\phi \leq 0.1$ and diverge at $\phi \geq 1.0$. At $\phi = 0.1$, MV coupling with MomentEncoder reduces Y_0 from 0.381 to 0.141 (no coupling vs moments), while DeepSets gives $Y_0 = 0.115$, confirming the coupling is active even in the stability-sweep regime.

5.8 Causal and Consistency Checks

Placebo test: shuffled embeddings. To confirm that the learned sensitivity is genuinely driven by population structure (not extra model capacity), we destroy the semantic link between the population and its embedding while preserving architecture and dimensionality (Table 7).

Shuffled and random embeddings all produce $h \approx 0.44$–0.47, similar to the narrow (low-competition) baseline. They do *not* produce $h = 0.10$ (the wide-population value). The low h under wide populations is therefore genuinely caused by the distributional signal, not by random embedding noise or extra model capacity. Since the architecture, dimensionality, and training procedure are unchanged, the collapse of the effect under shuffled and synthetic embeddings isolates the semantic content of the law representation as the sole driver.

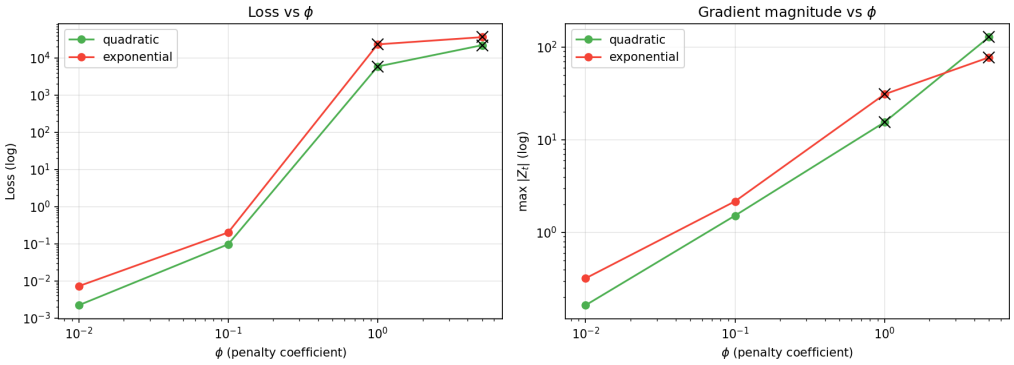

Figure 6: Loss and gradient magnitude vs ϕ for quadratic and exponential penalties. Sharp transition at $\phi \approx 1$.

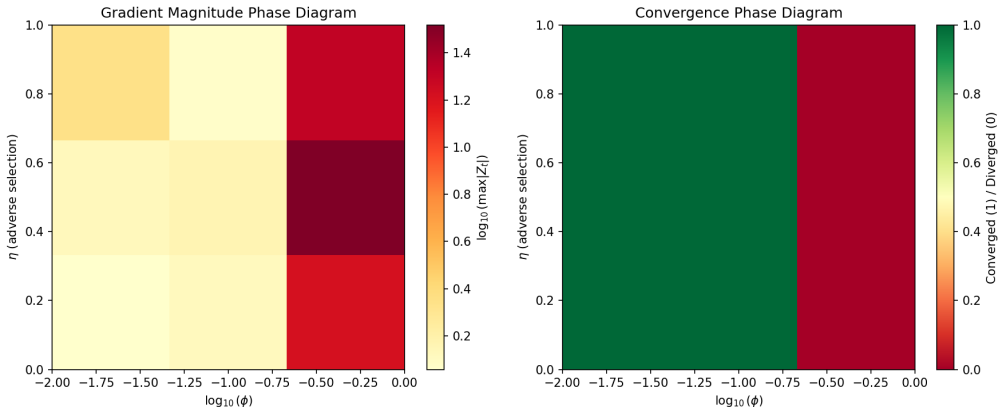

Figure 7: Phase diagram: $\phi \times \eta$. Left: gradient magnitude (log scale). Right: convergence map. Stability is determined by ϕ, not η.

Table 7: Placebo test: h under real vs destroyed embeddings.

Embedding source	h value	Interpretation
Real narrow (std 0.1)	0.434	Genuine signal
Real wide (std 5.0)	0.100	Genuine signal
Shuffled narrow (5 permutations)	0.44–0.45	$\approx$ narrow
Random Gaussian (5 draws)	0.45–0.47	$\approx$ narrow

Swap test. Feeding the narrow embedding to an agent at $q = 0$ gives $\delta_a = 0.66$ (conservative); feeding the wide embedding gives $\delta_a = 0.30$ (aggressive). Same agent state, different population embedding $\rightarrow$ different policy. This confirms the control pathway is causally active.

FD consistency at exogenous h. As a policy-level consistency check, we solve the single-agent FD problem with exogenous execution scaling $\lambda_{\text{eff}} = h \cdot \lambda$:

h (exogenous)	FD $V(0,0)$	FD spread
1.00	0.384	1.644
0.45 ($\approx$ narrow)	0.171	1.674
0.28 ($\approx$ medium)	0.106	1.685
0.11 ($\approx$ wide)	0.042	1.696

The FD values are directionally consistent: lower $h \to$ lower value, matching the neural network's learned ordering. FD spread barely changes with h (1.64–1.70), consistent with the discrete HJB structure where quotes depend on value increments rather than execution rate levels. The neural network's larger quote variation arises from the law embedding entering the subnet directly (two-stream architecture), providing an additional pathway for law-dependent variation beyond the standard HJB sensitivity to execution rates.

6 Discussion

Representation determines learnability. The central finding is that mean-field effects are *present in the model* but *invisible to the solver* unless the law encoder exposes the relevant distributional features. This has implications beyond this specific problem: any deep learning system that conditions on shared population statistics via broadcast features must account for normalisation layer interactions.

Architectural lessons. BatchNorm, while critical for Deep BSDE training stability Han et al. [2018], is fundamentally incompatible with broadcast conditioning signals. The two-stream architecture (BN on per-agent features, no BN on shared features) resolves this cleanly. FiLM-style modulation Perez et al. [2018] would be a natural alternative. More broadly, these findings suggest that any deep learning system conditioning on shared population statistics must carefully consider representation and normalisation, as architectural choices can mask genuine interaction effects.

Two pathways for law dependence. The current architecture provides two channels through which μ_t affects control: (i) the competitive factor $h(\Phi(\mu_t))$ entering the generator via execution rates, and (ii) the law embedding entering the subnet directly via the two-stream architecture. We have not disentangled how much of the observed policy variation comes from each. The FD consistency check validates the h-channel (value ordering matches exogenous scaling), but the larger quote shifts may partly reflect the direct subnet conditioning. Isolating each pathway's contribution — e.g., by ablating the law stream while keeping h, or vice versa — is an important direction for future work.

Economic interpretation. The learned competitive factor $h(\Phi(\mu_t))$ maps wider inventory distributions to lower fill rates, consistent with the intuition that dispersed populations contain more aggressive competitors. The resulting policy adaptations (narrower ask under competition, wider bid) are economically rational inventory management responses. We emphasise that this is a learned mapping, not a derived equilibrium condition: the economic story is plausible but not uniquely identified from the current evidence.

7 Conclusion

We have shown that deep BSDE solvers can be made sensitive to population structure in market-making, producing economically meaningful law-conditioned controls, but only when three architectural conditions are met: the law embedding enters the BSDE generator, broadcast features bypass BatchNorm, and the encoder captures distributional variance. When all three are satisfied, the solver learns distribution-dependent controls that disappear under semantic-destruction placebos and align directionally with finite-difference comparative statics under exogenous competition.

The deeper lesson is methodological: null results in mean-field deep learning may reflect architectural failures rather than genuine model invariance. Diagnosing the failure mode — cou-

pling pathway, normalisation, or representation — is essential before concluding that population structure does not matter.

Limitations. We demonstrate learned sensitivity to population structure, not equilibrium-correct sensitivity: the competitive factor network learns a monotone mapping (wider $\rightarrow$ lower h) but we have not verified that this corresponds to a Nash equilibrium fixed point or that fictitious play converges. Multi-seed results (Table 5) confirm the h-gap is robust (0.259 ± 0.055), but the Y_0 magnitude has higher variance across seeds (± 0.039). The competitive factor floor at $h = 0.1$ compresses the wide-population regime and may affect reported gap magnitudes. The adverse selection model uses a simple EMA signal. The experiments are at $d = 2$–3. We leave out-of-distribution generalization across unseen population shapes to future work.

Future work. Fictitious play convergence verification; FiLM conditioning as alternative to two-stream; DeepSets with variance-sensitive features (q^2 pooling); common noise via deep signatures Hu et al. [2025].

Code. `https://github.com/cgarryZA/DeepMVBSDEJ`

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